

Cisco Packet Tracer

Level 2: VLANs and Trunking

Cisco Packet Tracer | Topology: 2 Switches + 12 PCs across 3 departments

Overview

This section covers VLANs (Virtual Local Area Networks) and Trunking. Two fundamental concepts in enterprise networking. In a real company, different departments like HR, IT, and Finance all share the same physical network infrastructure, but they shouldn't be able to freely access each other's traffic. VLANs solve this by logically separating the network into isolated segments, even though everything is physically connected to the same switches.

Think of VLANs like separate floors in an office building. Everyone is in the same building (same physical network) but each floor (VLAN) operates independently, and people can't just walk into each other's floor without going through the right channels.

Network Topology

The topology for this lab simulates a small enterprise with two switches serving three departments across two sides of a building.

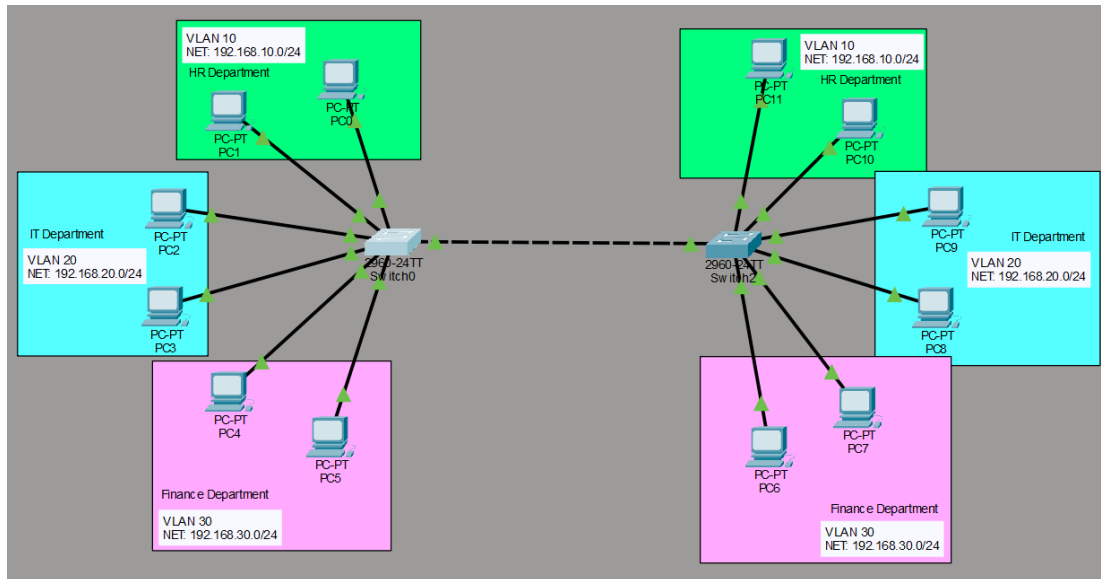


Figure 1. Network Topology

- **2 switches** – Cisco 2960-24TT (Switch0 and Switch2), connected to each other via trunk link
- **12 PCs** – 2 PCs per department per switch (6 per switch)
- **3 VLANs:**
 - **VLAN 10** – HR Department (192.168.10.0/24)
 - **VLAN 20** – IT Department (192.168.20.0/24)
 - **VLAN 30** – Finance Department (192.168.30.0/24)

NOTE: Devices in the same VLAN can communicate with each other. Devices in different VLANs cannot communicate by default. It requires a router or Layer 3 switch. Which will be covered in the next level

Configuration Steps

- **Step 1:** Design the network topology
- **Step 2:** Assign IP addresses to all host devices
- **Step 3:** Create VLANs on each switch and name them
- **Step 4:** Assign switch ports to their correct VLANs
- **Step 5:** Configure the trunk link between the two switches
- **Step 6:** Test VLAN Communication

Step 1: Design the Network Topology

The first step is to design and build the network layout in Packet Tracer. The Topology uses two Cisco 2960-24TT switches connected to each other, with each switch serving three departments – HR, IT, and Finance.

Each department is assigned a different VLAN and a different subnet, so their traffic stays completely isolated from each other at the network level. Color coding and labels are added in Packet Tracer to make the topology easier to read and understand at a glance.

Note: *using color coding and labels in your topology is good documentation practice. It makes it immediately clear which devices belong to which department and VLAN without having to trace every cable manually.*

Step 2: Assign IP addresses to Host devices

Each PC needs to be assigned a static IP address that matches its VLAN subnet. Click on the PC, go to the Desktop tab, click IP Configuration, and enter the IP address and subnet mask manually.

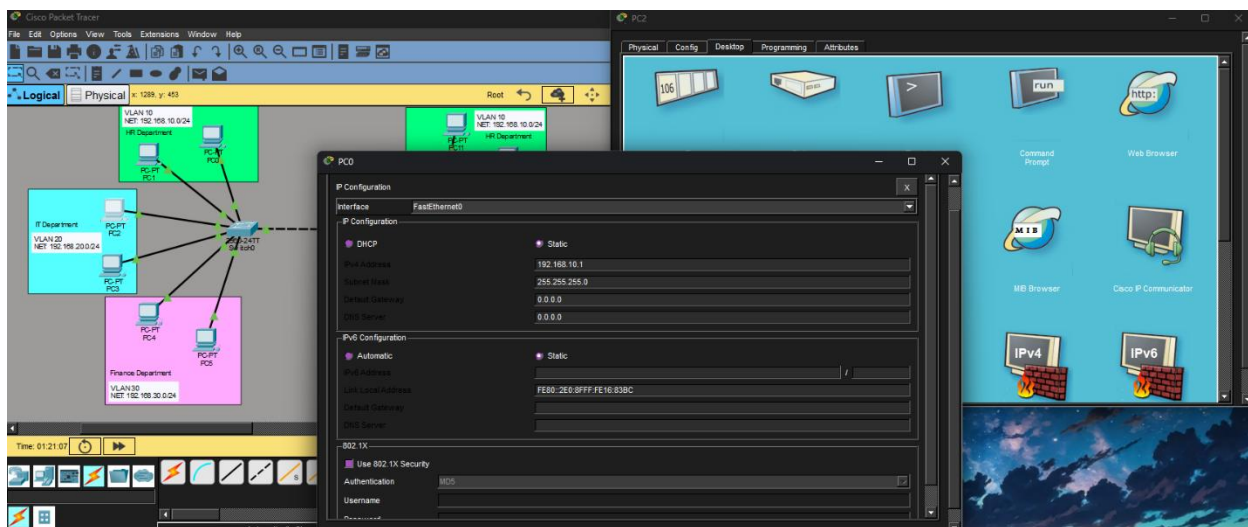


Figure 2. Assigning IP addresses to host devices

Understanding the IP Address Structure

The IP addresses used in this lab follow the format, 192.168.x.x, which is a private IP range, meaning these addresses only exist within the local network and are not routable on the public internet.

The 3rd octet (192.168.**x.x**) is used to identify which VLAN subnet a device belongs to:

- **192.168.10.x** – VLAN 10 (HR Department)
- **192.168.20.x** – VLAN 20 (IT Department)
- **192.169.30.x** – VLAN 30 (Finance Department)

By changing the 3rd octet, each VLAN gets its own separate subnet. Devices on different subnets cannot communicate with each other without a router – which is exactly the isolation VLANs are designed to achieve.

Note: *Two devices can share the same last octet (like .1 or .2) as long as they are on a different VLAN subnet. For example, 192.168.10.1 (HR) and 192.168.20.1 (IT) are separate addresses even though both end in .1. Both are on different networks entirely.*

IP Address Assignment Table

Based on the topology, here is how the IP addresses are distributed across all devices:

- **VLAN 10 – HR Department (192.168.10.0/24):**
 - PC0: 192.168.10.1 | PC1: 192.168.10.2 (Switch0 side)
 - PC10: 192.168.10.3 | PC11: 192.168.10.4 (Switch2 side)
- **VLAN 20 – IT Department (192.168.20.0/24):**
 - PC2: 192.168.20.1 | PC3: 192.168.20.2 (Switch0 side)
 - PC8: 192.168.20.3 | PC9: 192.168.20.4 (Switch2 side)
- **VLAN 30 – Finance Department (192.168.30.0/24):**
 - PC4: 192.168.30.1 | PC5: 192.168.30.2 (Switch0 side)
 - PC6: 192.168.30.3 | PC7: 192.168.30.4 (Switch0 side)

Step 3: Create VLANs on the Switch

With the IP addresses assigned, the next step is to create the VLANs on each switch through the CLI (Command Line Interface). Creating a VLAN on a switch tells it that this virtual network exists. Without this step, the switch doesn't know what VLAN 10, 20, or 30 are.

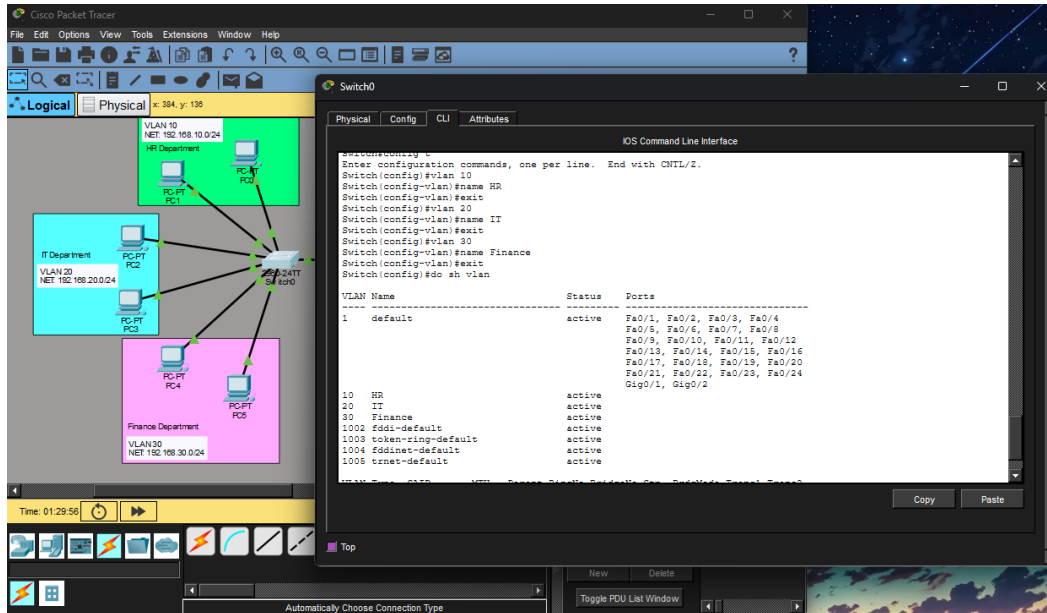


Figure 3. Creating VLAN on the Switch

Click on Switch0 and go to the CLI tab. Enter the following Commands:

Enter Privileged and Configuration Mode

en

Enables Privileged EXEC mode. Gives you elevated access to view and manages the switch. Think of this as unlocking admin level access.

config t

Enters Global Configuration mode. This is where you can make changes to the switch settings such as creating VLANs, assigning ports, setting hostnames, and more.

Create and name the VLANs

vlan 10

Creates VLAN 10 and moves you into VLAN configuration mode for that specific VLAN.

name HR

Assigns the name HR to VLAN 10. Naming VLANs makes them easier to identify when managing the switch later.

exit

Exits VLAN configuration mode and returns to Global Configuration mode.

Repeat the same steps for VLAN 20 (IT) and VLAN 30 (Finance):

vlan 20

name IT

exit

vlan 30

name Finance

exit

Note: *Make sure to do these steps as well for Switch2.*

Verify the VLANs

To confirm the VLANs were created and name correctly, run:

do sh vlan

do show vlan brief

This displays all VLANs configured on the switch along with their names and assigned ports. At this point the VLANs will show as created but with no ports assigned yet. That happens in the next step.

Note: *Cisco IOS accepts abbreviated commands as long as they are unique enough for the switch to understand. Most network engineers use the shorthand version for speed, but the full version is clearer when reading the documentation.*

Best Practice: *The 'do' prefix lets you run show command from inside configuration mode without having to exit first. Very useful when you want to verify changes without interrupting your configuration flow.*

Step 4: Assign Switch Ports to VLANs

Creating VLANs alone doesn't do anything. You need to tell the switch which physical ports belong to which VLAN. This is done by identifying which ports department PCs are connected to and assigning them accordingly.

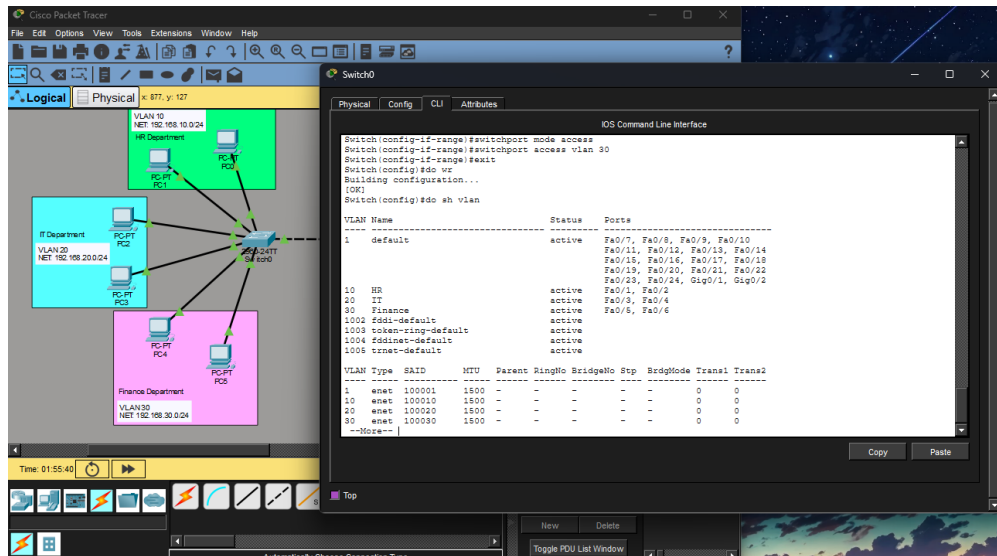


Figure 4. Assigning switch ports to VLAN

Identifying the Switch Ports

To find out which port a PC is connected to, hover your cursor over the green triangle (connection indicator) on the PC or the switch in Packet Tracer. It will show the port name. For example, Fa01/ or Fa0/2.

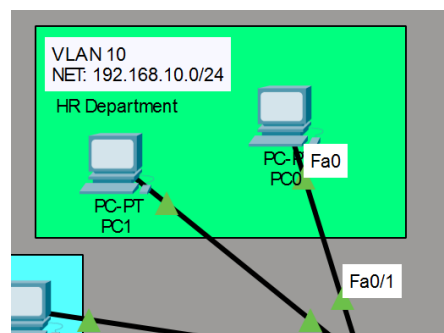


Figure 5. Hovering green indicator to see port name.

Based on the topology, here are the port assignments for Switch0.

- **VLAN 10 (HR):** Fa0/1, Fa0/2
- **VLAN 20 (IT):** Fa0/3, Fa0/4
- **VLAN 30 (Finance):** Fa0/5, Fa0/6

Configure Access ports

Go to Switch0 and enter configuration mode. Then assign each group of ports to their VLAN:

int range fa0/1-2

Select both ports Fa0/1 and Fa0/2 at the same time so you can configure them together instead of one by one.

switchport mode access

Sets the port to Access mode. This means the port only belongs to one VLAN and connects to an end device like a PC or printer. Access mode tells the switch that whatever is connected to this port is a regular device, not another switch. Traffic coming in and out of this port carries only traffic of the assigned VLAN, nothing else.

switchport access vlan 10

Assigns these ports to VLAN 10. Any device connected to Fa0/1 or Fa0/2 is now part of the HR network.

exit

Repeat the same process for VLAN 20 and VLAN 30:

int range fa0/3-4

switchport mode access

switchport access vlan 20

exit

int range fa0/5-6

switchport mode access

switchport access vlan 30

exit

do wr

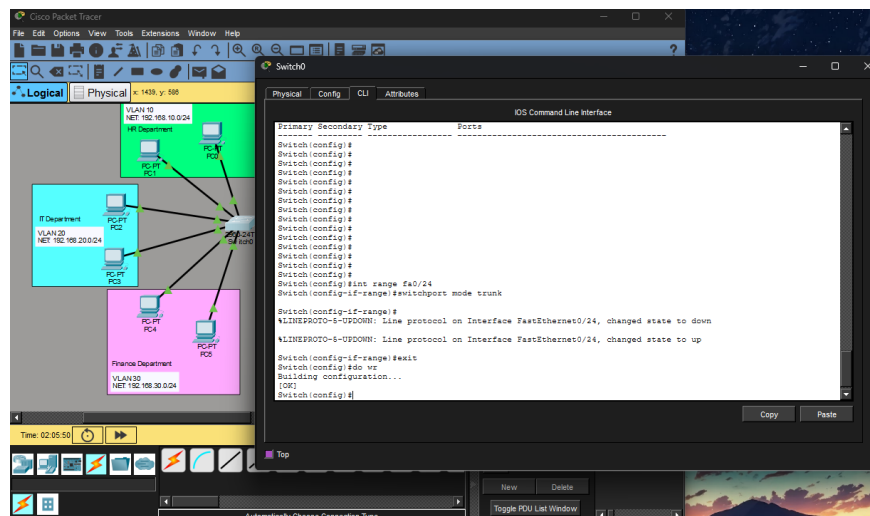
Saves all current configuration changes to the switch's memory so they persist if the device is restarted.

Run

again, to verify that the ports are now showing under their correct VLANs. Previously all ports defaulted to VLAN 1. After this step, they should now be assigned to VLAN 10, 20, and 30 respectively.

Important: Repeat this entire Step 4 process to Switch 2 as well. Every switch needs to have its ports assigned to the correct VLANs independently.

With VLANs configured on both switches, the final configuration step is to set up the trunk link between them. Without trunking, each switch operates in its own isolated world and the VLANs on Switch0 have no way to reach the same VLANs on Switch2.



A trunk link is a connection between two switches that carries from multiple VLANs simultaneously. Think of it like a multi-lane highway between two buildings. HR, IT, and Finance traffic all travel on the same physical cable but stay in their own separate lanes (VLANs).

Based on the topology, the two switches are connected via Fa0/24. Click on Switch0, go to CLI, and enter:

int fa0/24

switchport mode trunk

exit

do wr

This sets port Fa0/24 on Switch0 to Trunk mode. It will now carry traffic from all VLANs between the two switches.

Important: Repeat the same trunk configuration on Switch2's Fa0/24 port. Both ends of the trunk link must be set to trunk mode. If only one side is configured, the link won't work correctly.

Note: Unlike access ports which carry only one VLAN, trunk ports carry all VLANs by default. This is what allows HR devices on Switch0 to communicate with HR devices on Switch2, while still keeping them isolated from IT and Finance.

Step 6: Testing VLAN Communication

With everything configured, the last step is to verify the VLANs are working correctly. Devices in the same VLAN should communicate, and devices in different VLANs should not.

Test 1 – Same VLAN Communication

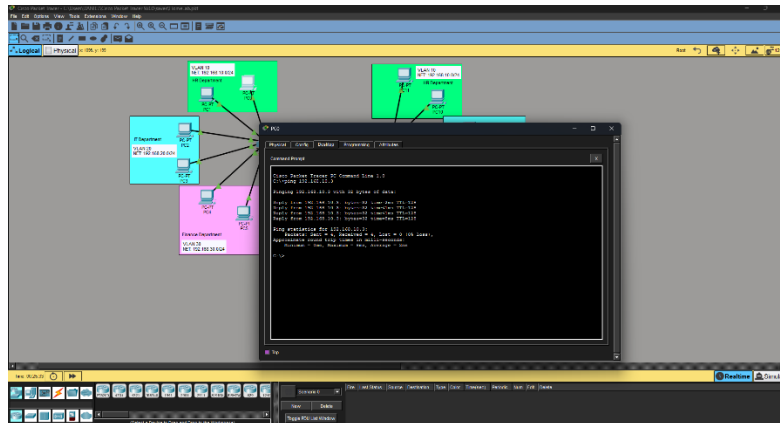


Figure 7. Ping reply received from 192.168.10.3

Click on PC0 (HR, Switch0 side) and go to Desktop > Command Prompt. Ping PC10 or PC11 which are also in VLAN 10 on Switch2:

ping 192.168.10.3

Expected result: Reply received – HR devices can communicate across both switches through the trunk link.

Test 2 – Different VLAN Communication

Now try pinging a device in a different VLAN – for example an IT department PC:

ping 192.168.20.1

Expected result: Request timed out – HR and IT are on different VLANs and cannot communicate without a router. This confirms that VLAN isolation is working correctly.

Note: Getting a 'Request timed out' when pinging across VLANs is the correct and expected behavior at this stage. It confirms that the VLAN segmentation is working as intended. Making VLANs talk to each other requires inter-VLAN routing which is covered in the next level.

Key Takeaways

- **VLANs logically separate a network** – even though all devices share the same physical switches, VLANs keep their traffic completely isolated from each other
- **The third octet identifies the VLAN subnet** – 192.168.10.x for HR, 192.168.20.x for IT, 192.168.30.x for Finance keeps addressing organized and easy to manage
- **Access ports connect to end devices** – they carry traffic for one VLAN only and are used for PCs, printers, and other end devices.
- **Trunk ports connect switches together** – they carry traffic from all VLANs simultaneously and must be configured on both ends of the link
- **Always save your configuration** – running 'do wr' after changes ensures your configuration survives a restart
- **VLANs cannot communicate by default** – inter-VLAN routing requires a router or Layer 3 switch, which is the next step in the lab series
- **Both switches need to be configured** – VLANs and trunk ports must be set up independently on every switch in the network

